THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

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Software Engineering and Project management Assignment - 1

Q1 Software process models

→ Software process models or development models are various processes or methods that are chosen for project development depending on the objectives and goals of the project. Many development life cycle models have been developed to achieve various essential objectives. Models specify the various steps of the process and in order in which they are executed.

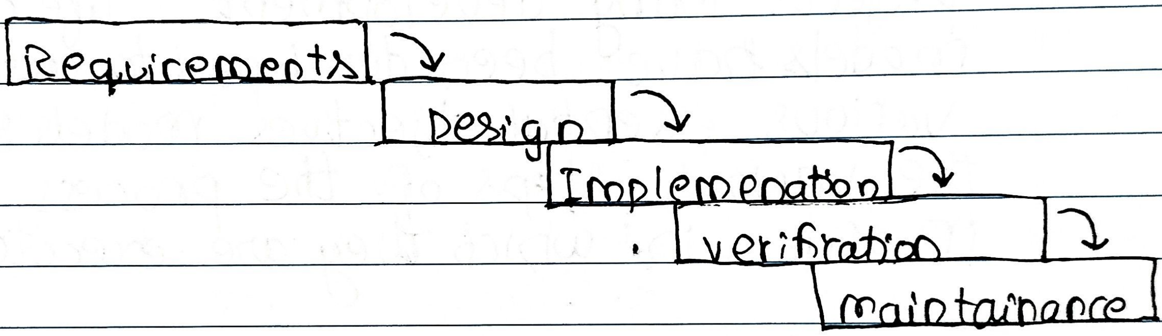
Types of model

→ waterfall model
→ V-model
→ agile model
→ incremental model
→ RAD model
→ Iterative model
→ spiral model
→ Prototype model

1. Waterfall model

→ waterfall model is a linear and sequencial model, which means that a development phase cannot begin until the previous phase is completed. We cannot overlap phases in waterfall model.

Diagram

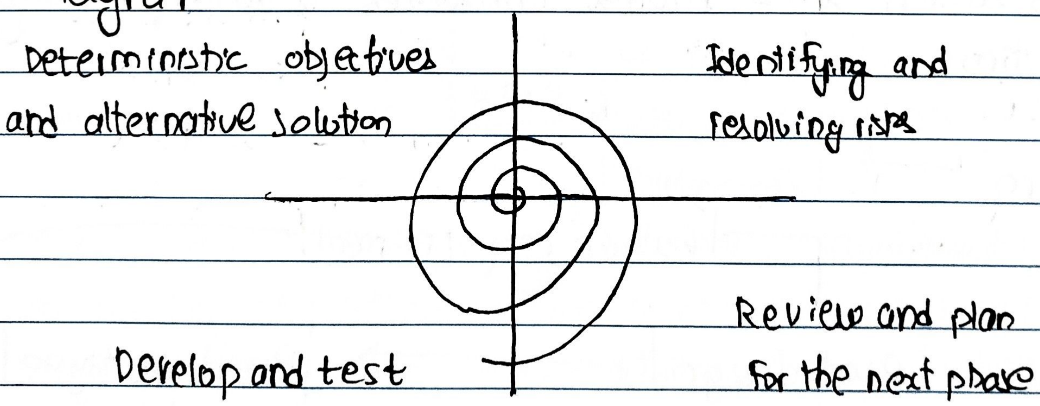


Similarly waterfall model also works, once one phase of development is completed then we move to the next Phase but cannot go back to the previous phase. In the waterfall model, the output of one phase servers as the input for the other phase.

2. Spiral model

→ This type of model is a software development process model. This model has characteristics of both iterative and waterfall models. This model is used in projects which are large and complex. This model was named spiral because if one looks at its figure it looks like a spiral, in which a long curved line starts from the center point and makes many loops around it. The number of loops in the spiral is not decided in advance but it depends on the size of the project and the changing requirements of the users. We also call each loop of the spiral a phase of the software development process.

Diagram



In spiral model the entire process of software development is described in four phases which are repeated until the project is completed.

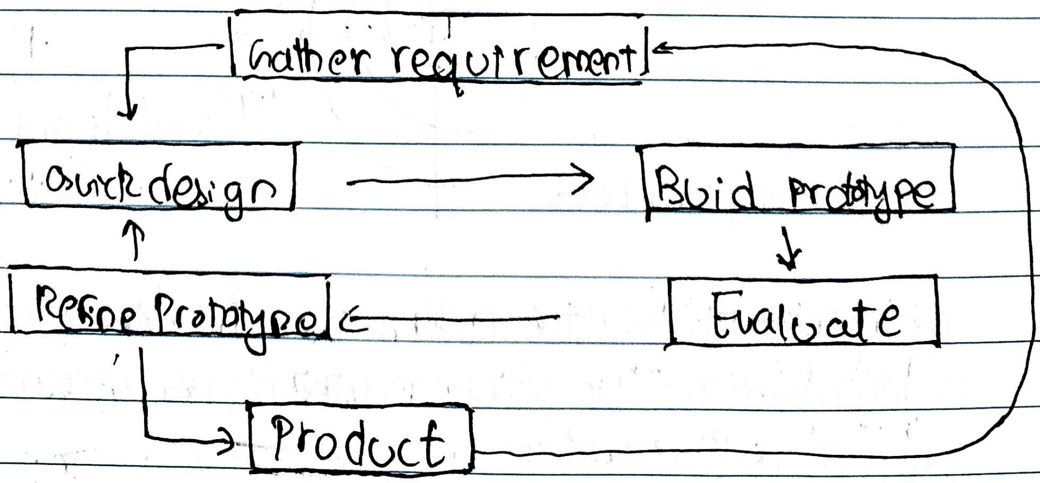
A software project goes through these loops again and again in iterations. After each iteration a more and more complete version of the software is developed. The most special thing about this model is that risks are identified in each phase and they are resolved through prototyping. This feature is also called risk handling.

since it also includes the approaches of other SDLC models. This also called meta model

Prototype model

→ Prototype model is an activity in which prototypes of software applications are created. First a prototype is created and then the final product is manufactured based on that prototype. one problem in the model is that if the end users are not satisfied with the prototype model. then ~~a new~~ new prototype model is created again. due to which this model consumes a lot of money and time

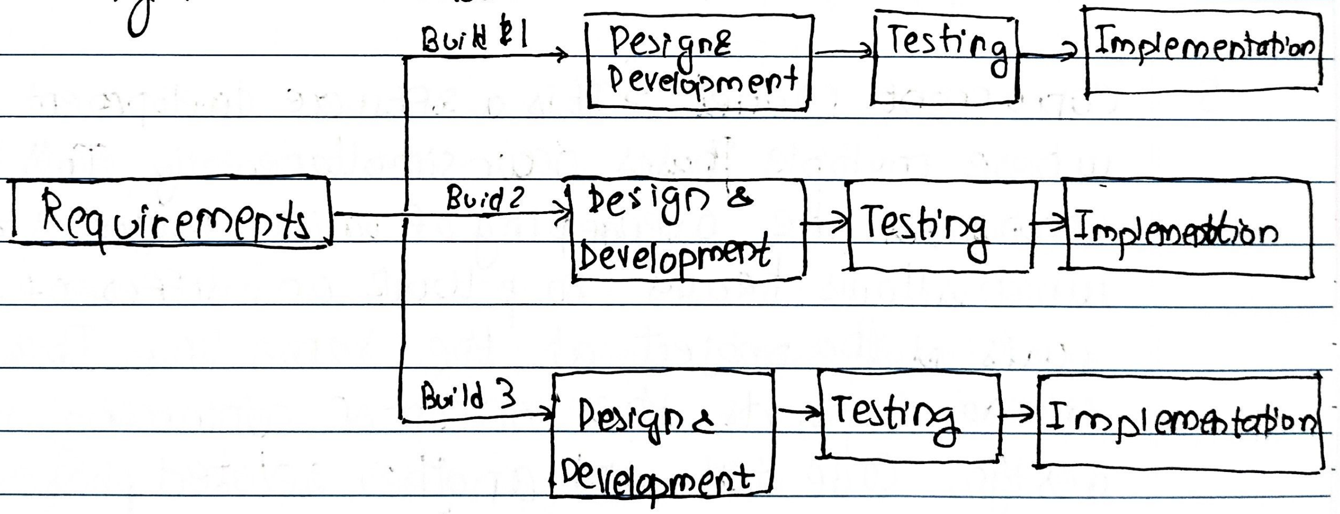
Diagram



The prototype model was developed to overcome the shortcomings of the waterfall model. This model is created when we do not know the requirements well. The specialty of this model is that this model can be used with other models as well as alone.

4. Incremental model

→ In this model, the software development Process is divided into several increments and the same phases are followed in each increment. In simple language, under this model a complex project is developed in many modules or builds



For example: we collect the customer's requirements. now instead of making the entire software at once, we first take some requirements and based on them create a module or function of the software and deliver it to the customer. Then we take some more requirements and based on them add another module to the that software.

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→ similarly, modules are added to the software in each increment until the complete system is created. However, the requirements for making a complex project in multiple iterations/parts should be clear.

→ The entire principle of incremental methodology it starts by developing an initial implementation, then user feedback is taken on it and it's developed through several versions until an accepted system is developed. Important functionalities of the software are developed in the initial iterations.

→ Each subsequent release of a software module adds functions to the previous release. This process continues until the final software is obtained.

5. Concurrent model → It is a software development approach where multiple phases occur simultaneously. This model emphasizes the overlapping of various project stages. which allows teams to work on different parts of the project at the same time. This model supports the concept of concurrency is that design, code, test, and another related phase occur simultaneously.

They include minimizing development time, encouraging effective interaction between development teams and increasing the versatility of the resulting product due to the possibility of feedback at any step of the cycle.

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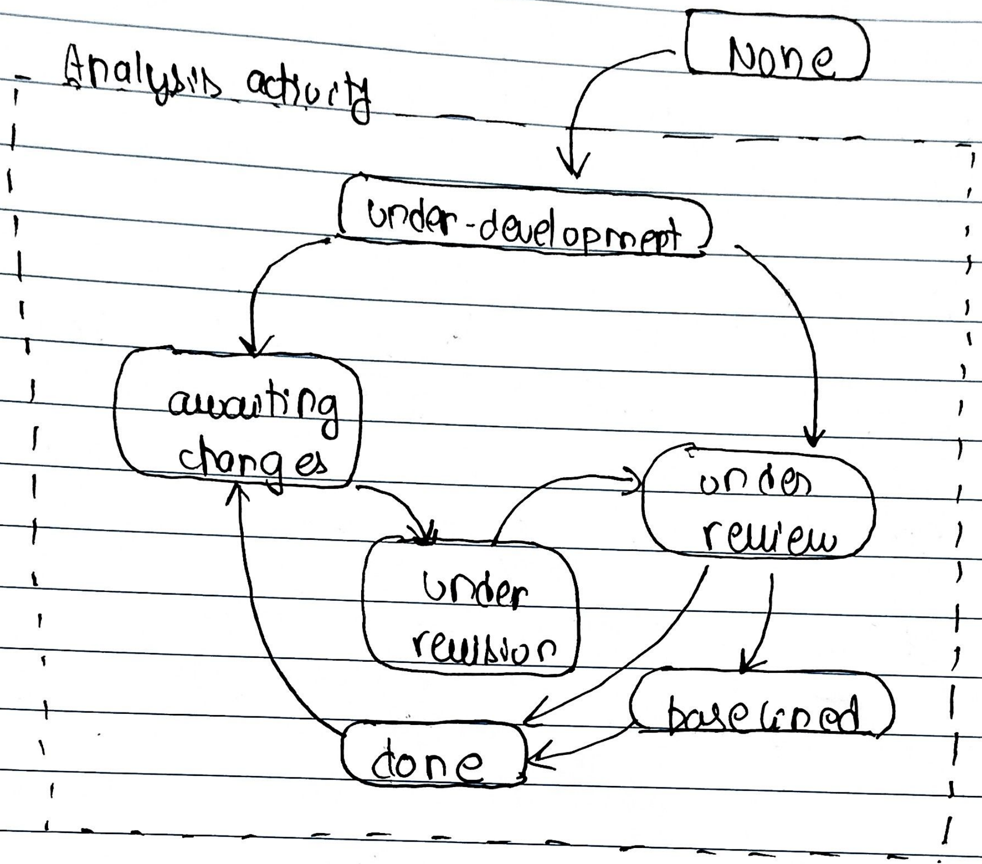
Diagram



Features of the concurrent model

→ Parallel workflow: Several phases start and run concurrently like the design phase, development phase, and the testing phase allowing completion of a project more expediently.

→ Real time communication: Eg Everyone gets repeated feedback hence the team works and solutions together to ensure the common goal is achieved.

→ Flexibility: Stakeholders can incorporate new changes and improvements easily since one phase operates in tangent to another.